TRIPATHI STUDY ZONE

JEE MAIN | NEET | AIIMS | KVPY | BITSAT

Mock Test 5

Time : 3 hrs. **Max. Marks : 300**

INSTRUCTIONS

1. This test will be a 3 hours Test.
2. This test consists of Physics, Chemistry and Mathematics questions with equal weightage of 100 marks.
3. Each question is of 4 marks.
4. There are three parts in the question paper consisting of Physics (Q.no.1 to 30), Chemistry (Q.no.31 to 60) and Mathematics (Q. no.61 to 90). Each part is divided into two sections, Section A consists of 20 multiple choice questions & Section B consists of 10 Numerical value answer Questions.
   In Section B, candidates have to attempt **only 5 questions out of 10**.
5. There will be only one correct choice in the given four choices in Section A. For each question 4 marks will be awarded for correct choice, 1 mark will be deducted for incorrect choice and zero mark will be awarded for unattempted question. For Section B 4 marks will be awarded for correct answer and zero for marked for each review / unattempted/incorrect answer.
6. Any textual, printed or written material, mobile phones, calculator etc. is not allowed for the students appearing for the test.
7. All calculations / written work should be done in the rough sheet provided.

PHYSICS

Section - A

1. A lift is moving down with acceleration a. A man in the lift drops a ball inside the lift. The acceleration of the ball as observed by the man in the lift and a man standing stationary on the ground are respectively

   (1) g, g (2) $g-a, g-a$

   (3) $g-a, g$ (4) a, g

2. Relative permittivity and permeability of a material are ε_r and μ_r, respectively. Which of the following values of these quantities are allowed for a diamagnetic material?

   (1) $\varepsilon_r = 0.5, \mu_r = 1.5$ (2) $\varepsilon_r = 1.5, \mu_r = 0.5$

   (3) $\varepsilon_r = 0.5, \mu_r = 0.5$ (4) $\varepsilon_r = 1.5, \mu_r = 1.5$

3. Let there be a spherically symmetric charge distribution with charge density varying as $\rho(r) = \rho_0\left(\frac{5}{4} - \frac{r}{R}\right)$ upto $r = R$, and $\rho(r) = 0$ for $r > R$, where r is the distance from the origin. The electric field at a distance $r(r < R)$ from the origin is given by

Space for Rough Work

(1) $\frac{\rho_0 r}{4\varepsilon_0}\left(\frac{5}{3}-\frac{r}{R}\right)$ (2) $\frac{4\pi\rho_0 r}{3\varepsilon_0}\left(\frac{5}{3}-\frac{r}{R}\right)$

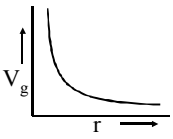
(3) $\frac{\rho_0 r}{4\varepsilon_0}\left(\frac{5}{4}-\frac{r}{R}\right)$ (4) $\frac{\rho_0 r}{3\varepsilon_0}\left(\frac{5}{4}-\frac{r}{R}\right)$

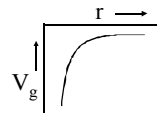
4. An ideal gas enclosed in a vertical cylindrical container supports a freely moving piston of mass M. The piston and the cylinder have equal cross sectional area A. When the piston is in equilibrium, the volume of the gas is V_0 and its pressure is P_0. The piston is slightly displaced from the equilibrium position and released. Assuming that the system is completely isolated from its surrounding, the piston executes a simple harmonic motion with frequency

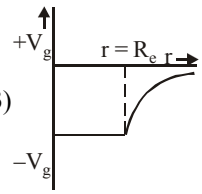
(1) $\frac{1}{2\pi}\frac{A\gamma P_0}{V_0 M}$ (2) $\frac{1}{2\pi}\frac{V_0 M P_0}{A^2\gamma}$

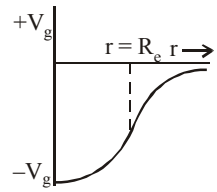
(3) $\frac{1}{2\pi}\sqrt{\frac{A^2\gamma P_0}{MV_0}}$ (4) $\frac{1}{2\pi}\sqrt{\frac{MV_0}{A\gamma P_0}}$

5. The correct graph between the gravitational potential (V_g) due to a hollow sphere and distance from its centre will be

(1) 

(2) 

(3) 

(4) 

6. In a cubical vessel are enclosed *n* molecules of a gas each having a mass *m* and an average speed *v*. If ℓ is the length of each edge of the cube, the pressure exerted by the gas will be

(1) $\frac{nmv^2}{\ell^3}$ (2) $\frac{nm^2v}{2\ell^3}$

(3) $\frac{mnv^2}{3\ell^3}$ (4) $\frac{nmv}{2\ell}$

7. A sphere, a cube and a thin circular plate all made of the same material and having the same mass, are initially heated to a temperature of 200°C. Which of these objects will cool slowest when left in air at room temperature?

(1) the sphere

(2) the cube

(3) the circular plate

(4) all will cool at same rate

8. An electromagnetic wave in vacuum has the electric and magnetic field $\vec{E}$ and $\vec{B}$, which are always perpendicular to each other. The direction of polarization is given by $\vec{X}$ and that of wave propagation by $\vec{k}$. Then

(1) $\vec{X} \parallel \vec{B}$ and $\vec{k} \parallel \vec{B}\times\vec{E}$

(2) $\vec{X} \parallel \vec{E}$ and $\vec{k} \parallel \vec{E}\times\vec{B}$

(3) $\vec{X} \parallel \vec{B}$ and $\vec{k} \parallel \vec{E}\times\vec{B}$

(4) $\vec{X} \parallel \vec{E}$ and $\vec{k} \parallel \vec{B}\times\vec{E}$

9. A particle of mass m_1 collides head-on with another stationary particle of mass m_2 ($m_2 > m_1$). The collision is perfectly inelastic. The fraction of kinetic energy which is converted into heat in this collision is

(1) $m_2/(m_1+m_2)$ (2) $m_1/(m_1+m_2)$

(3) $m_1/(m_1-m_2)$ (4) $m_2/(m_1-m_2)$

10. The instantaneous values of current and voltage in an A.C. circuit are $I = 4 \sin \omega t$ and $E = 100 \cos\left(\omega t + \frac{\pi}{3}\right)$ respectively. The phase difference between voltage and current is

(1) $\frac{\pi}{3}$ (2) $\frac{2\pi}{3}$

(3) $\frac{5\pi}{6}$ (4) $\frac{7\pi}{6}$

11. Which of the following units denotes the dimension $\frac{ML^2}{Q^2}$, where Q denotes the electric charge?

(1) Wb/m^2 (2) Henry (H)

(3) H/m^2 (4) Weber (Wb)

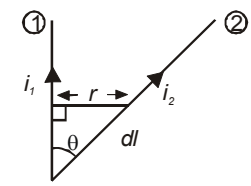
12. Wires 1 and 2 carrying currents i_1 and i_2 respectively are inclined at an angle θ to each other. What is the force on a small element dl of wire 2 at a distance of r from wire 1 (as shown in figure) due to the magnetic field of wire 1?

(1) $\frac{\mu_0}{2\pi r} i_1 i_2\, dl \tan\theta$

(2) $\frac{\mu_0}{2\pi r} i_1 i_2\, dl \sin\theta$

(3) $\frac{\mu_0}{2\pi r} i_1 i_2\, dl \cos\theta$

(4) $\frac{\mu_0}{4\pi r} i_1 i_2\, dl \sin\theta$



13. In Young's double slit experiment, one of the slit is wider than other, so that amplitude of the light from one slit is double of that other slit. If I_m be the maximum intensity, the resultant intensity I when they interfere at phase difference ϕ is given by :

(1) $\frac{I_m}{9}(4+5\cos\phi)$

(2) $\frac{I_m}{3}\left(1+2\cos^2\frac{\phi}{2}\right)$

(3) $\frac{I_m}{5}\left(1+4\cos^2\frac{\phi}{2}\right)$

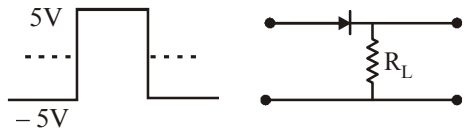
(4) $\frac{I_m}{9}\left(1+8\cos^2\frac{\phi}{2}\right)$

14. A pendulum consists of a wooden bob of mass m and length 'ℓ'. A bullet of mass m_1 is fired towards the pendulum with a speed v_1. The bullet emerges out of the bob with a speed $v_1/3$, and the bob just completes motion along a vertical circle. Then v_1 is

(1) $\left(\frac{m}{m_1}\right)\sqrt{5g\ell}$ (2) $\frac{3}{2}\left(\frac{m}{m_1}\right)\sqrt{5g\ell}$

(3) $\frac{2}{3}\left(\frac{m_1}{m}\right)\sqrt{5g\ell}$ (4) $\left(\frac{m_1}{m}\right)\sqrt{g\ell}$

15. In a *p-n* junction diode, a square input signal of 10 V is applied as shown in figure.



The output signal across R_L will be

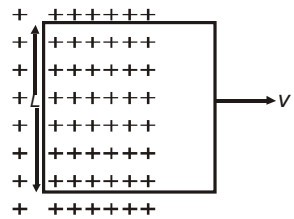
(1) 

(2) 

(3) 

(4) 

16. A conducting square loop of side *L* and resistance *R* moves in its plane with a uniform velocity v perpendicular to one of its sides. A magnetic induction *B* constant in time and space, pointing perpendicular and into the plane at the loop exists everywhere with half the loop outside the field, as shown in figure. The induced emf is



(1) zero (2) *RvB*

(3) *vBL/R* (4) *vBL*

17. Sinusoidal carrier voltage of frequency 1.5 MHz and amplitude 50 V is amplitude modulated by sinusoidal voltage of frequency 10 kHz producing 50% modulation. The lower and upper side-band frequencies in kHz are

(a) 1490, 1510 (b) 1510, 1490

(c) $\frac{1}{1490}, \frac{1}{1510}$ (d) $\frac{1}{1510}, \frac{1}{1490}$

18. A ray of light is incident at an angle of 60° on one face of a prism of angle 30°. The ray emerging out of the prism makes an angle of 30° with the incident ray. The emergent ray is

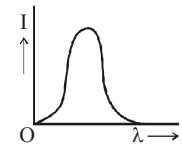
(1) Normal to the face through which it emerges

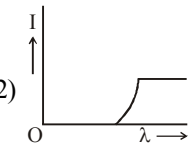
(2) Inclined at 30° to the face through which it emerges

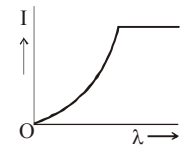
(3) Inclined at 60° to the face through which it emerges

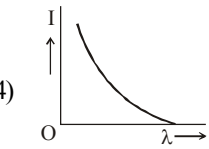
(4) Inclined at 90° to the normal at face through which it emerges

19. The anode voltage of a photocell is kept fixed. The wavelength λ of the light falling on the cathode is gradually changed. The plate current I of the photocell varies as follows :

(1) 

(2) 

(3) 

(4) 

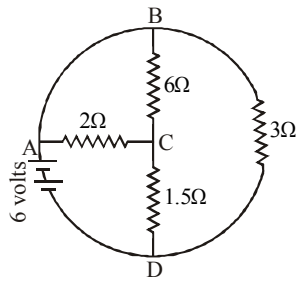
20. A proton, a deutron and an α particle accelerated through the same potential difference enter a region of uniform magnetic field, moving at right angles to B. What is the ratio of their K.E.?

(1) 1 : 2 : 2 (2) 2 : 2 : 1

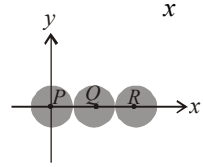
(3) 1 : 2 : 1 (4) 1 : 1 : 2

Section - B

21. In the circuit shown, the total current supplied by the battery is _____ A.



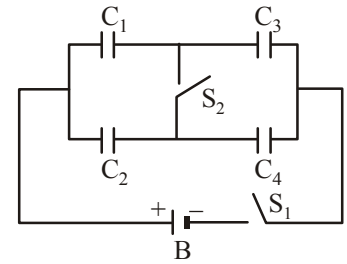
22. Three identical spheres, each of mass 1 kg are kept as shown in figure, touching each other, with their centres on a straight line. If their centres are marked *P, Q, R* respectively, the distance of centre of mass of the system from *P* is $\frac{PQ+PR}{x}$. Find the value of *x*.

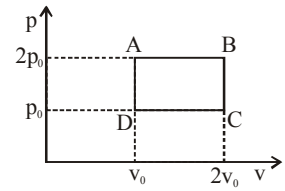


23. The half life of a radioactive substance is 20 minutes. The approximate time interval $(t_2 - t_1)$ between the time t_2 when $\frac{2}{3}$ of it had decayed and time t_1 when $\frac{1}{3}$ of it had decayed is _____ min.

24. A wooden block of volume 1000 cc is suspended from a spring balance. It weighs 12 N in air. It is then held suspended in water with half of it inside water. What would be the reading in spring balance (in N) now?

25. In the figure battery B supplies 12 V. Take $C_1 = 1$ μF, $C_2 = 2$ μF, $C_3 = 3$ μF, $C_4 = 4$ μF. Charge on capacitor C_1 when only S_1 is closed is _____ μC.



26. 

The above p-v diagram represents the thermodynamic cycle of an engine, operating with an ideal monatomic gas. The amount of heat, extracted from the source in a single cycle is *x* p_0v_0. Find the value of *x*.

27. A hydrogen-like atom has one electron revolving around a stationary nucleus. The energy required to excite the electron from the second orbit to the third orbit is 47.2 eV. The atomic number of the atom is _____.

28. A boy playing on the roof of a 10 m high building throws a ball with a speed of 10m/s at an angle of 30° with the horizontal. How far (in m) from the throwing point will the ball be at the height of 10 m from the ground ?

$[g = 10 \text{m/s}^2, \sin 30^o = \frac{1}{2}, \cos 30^o = \frac{\sqrt{3}}{2}]$

29. Two periodic waves of intensities I_1 and I_2 pass through a region at the same time in the same direction. The sum of the maximum and minimum intensities is x $(I_1 + I_2)$. Find the value of *x*.

30. A copper wire of length 1.0 m and a steel wire of length 0.5 m having equal cross-sectional areas are joined end to end. The composite wire is stretched by a certain load which stretches the copper wire by 1 mm. If the Young's modulii of copper and steel are respectively 1.0×10^{11} Nm^{-2} and 2.0×10^{11} Nm^{-2}, the total extension of the composite wire is _____ mm.

CHEMISTRY

Section - A

31. In which of the following arrangements, the sequence is not strictly according to the property written against it?
    (1) $CO_2 < SiO_2 < SnO_2 < PbO_2$: increasing oxidising power
    (2) $NH_3 < PH_3 < AsH_3 < SbH_3$: increasing basic strength
    (3) HF < HCl < HBr < HI: increasing acid strength
    (4) B < C < O < N: increasing first ionisation enthalpy.

32. Aluminothermy used for on the spot welding of large iron structure is based on the fact that-
    (1) As compared to iron, aluminium has greater affinity for oxygen.
    (2) As compared to aluminium, iron has greater affinity for oxygen.
    (3) Reaction between aluminium and oxygen is endothermic.
    (4) Reaction between iron and oxygen is endothermic.

33. The charge/size ratio of a cation determines its polarizing power. Which one of the following sequences represents the increasing order of the polarizing power of the cationic species, K^+, Ca^{2+}, Mg^{2+}, Be^{2+}?
    (1) $Ca^{2+} < Mg^{2+} < Be^{+} < K^{+}$
    (2) $Mg^{2+} < Be^{2+} < K^{+} < Ca^{2+}$
    (3) $Be^{2+} < K^{+} < Ca^{2+} < Mg^{2+}$
    (4) $K^{+} < Ca^{2+} < Mg^{2+} < Be^{2+}$.

34. A 1.0 M solution with respect to each of the metal halides AX_3, BX_2, CX_3 and DX_2 is electrolysed using platinum electrodes. If

$E^\circ_{A^{3+}/A} = 1.50$ V, $E^\circ_{B^{2+}/B} = 0.3$ V,

$E^\circ_{C^{3+}/C} = -0.74$ V, $E^\circ_{D^{2+}/D} = -2.37$ V.

The correct sequence in which the various metals are deposited at the cathode is :
    (1) A, B, C, D (2) A, B, C
    (3) D, C, B, A (4) C, B, A

35. An ideal gas undergoes isothermal expansion at constant pressure. During the process :
    (1) enthalpy increases but entropy decreases.
    (2) enthalpy remains constant but entropy increases.
    (3) enthalpy decreases but entropy increases.
    (4) Both enthalpy and entropy remain constant.

36. Among the reactions given below for B_2H_6, the one which does not take place is

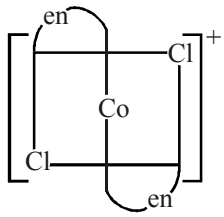
(1) $B_2H_6 + HCl \longrightarrow B_2H_5Cl + H_2$

(2) $2B_2H_6 + 6NH_3 \xrightarrow{\Delta} B_3N_3H_6$ (borazine)

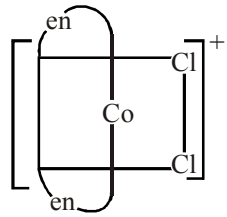
(3) $B_2H_6 + 2N(CH_3)_3 \longrightarrow 2(CH_3)_3NBH_3$

(4) $B_2H_6 + 6C_2H_4 \xrightarrow{H_3O^+} 3C_2H_5OH + 2B(OH)_3$

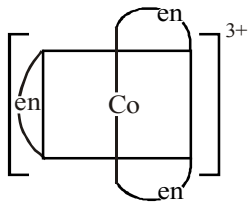
37. Which of the following ions are optically active?



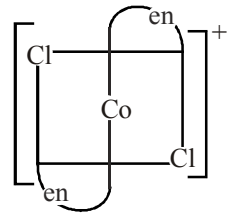
I



II



III



IV

(1) I only (2) II only

(3) II and III (4) IV only

38. The pair of compounds having metals in highest oxidation state is:

(1) MnO_2 and CrO_2Cl_2

(2) $[NiCl_4]^{2-}$ and $[CoCl_4]^{2-}$

(3) $[Fe(CN)_6]^{3-}$ and $[Cu(CN)_2]^{2-}$

(4) $[FeCl_4]^-$ and Co_2O_3

39. A metal gives two chlorides A and B. A gives black precipitate with NH_4OH and B gives white. With KI, B gives a red precipitate soluble in excess of KI. A and B are respectively :

(1) $HgCl_2$ and Hg_2Cl_2 (2) Hg_2Cl_2 and $HgCl_2$

(3) $HgCl_2$ and $ZnCl_2$ (4) $ZnCl_2$ and $HgCl_{24}$

40. In which reaction, there is change in oxidation number of N

(1) $2NO_2 \rightarrow N_2O_4$

(2) $NH_4OH \rightarrow NH_4^+ + OH^-$

(3) $N_2O_5 + H_2O \rightarrow 2HNO_3$

(4) $2NO_2 + H_2O \rightarrow HNO_3 + HNO_2$

41. Potassium permanganate acts as an oxidant in neutral, alkaline as well as acidic media. The final products obtained from it in the three conditions are, respectively

(1) MnO_4^{2-}, Mn^{3+} and Mn^{2+}

(2) MnO_2, MnO_2 and Mn^{2+}

(3) MnO_2, MnO_2^+ and Mn^{3+}

(4) MnO, MnO_2 and Mn^{2+}

42. An organic compound 'X' on ozonolysis followed by reduction with Zn/H_2O gives 2 moles of $H-\overset{O}{\overset{\|}{C}}-H$ and $H-\overset{O}{\overset{\|}{C}}-CH_2-\overset{O}{\overset{\|}{C}}-H$. 'X' is

(1) $CH_2=CH-CH_2-CH=CH_2$

(2) $CH_2=CH-CH_2-CH_2-CH=CH_2$

(3) $H-C\equiv C-C\equiv C-H$

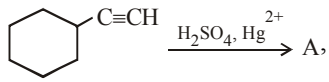
(4) $CH_2=CH-CH=CH_2$

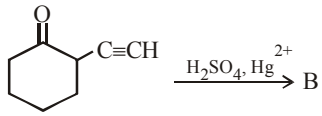
43. Which one of the following compounds is resistant to nucleophilic attack by hydroxyl ions?
(1) Methyl acetate (2) Acetonitrile
(3) Diethyl ether (4) Acetamide

44. To detect iodine in presence of bromine, the sodium extract is treated with $NaNO_2$ + glacial acetic acid + CCl_4. Iodine is detected by the appearance of
(1) yellow colour of CCl_4 layer.
(2) purple colour of CCl_4.
(3) brown colour in the organic layer of CCl_4.
(4) deep blue colour in CCl_4.

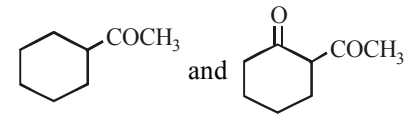
45. An organic compound A ($C_4H_{10}O$) has two enantiomeric forms and on dehydration it gives B(major product) and C (minor product). B and C are treated with HBr/ Peroxide and the compounds so produced were subjected to alkaline hydrolysis then-
(1) B will give an isomer of A
(2) C will give an isomer of A
(3) Neither of them will give isomer of A
(4) Both B and C will give isomer of A

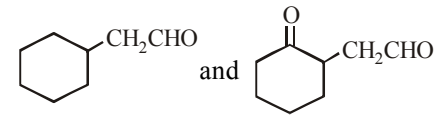
46. In which of the following polymers, empirical formula resembles with monomer?
(1) Bakelite (2) Teflon
(3) Nylon-6, 6 (4) Dacron

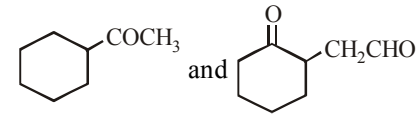
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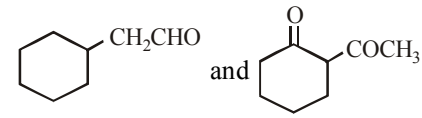


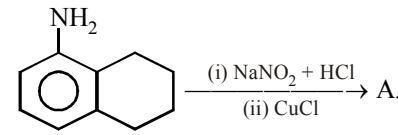
The respective compounds A and B are

(1) 

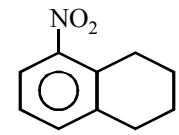
(2) 

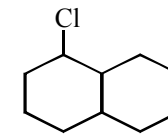
(3) 

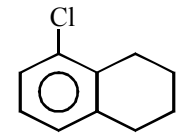
(4) 

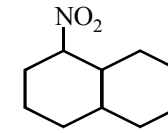
48. 

Compound 'A' is

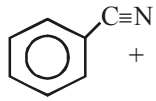
(1) 

(2) 

(3) 

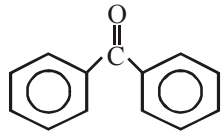
(4) 

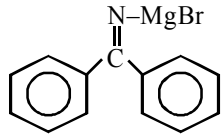
49. Allyl phenyl ether can be prepared by heating:
(1) $C_6H_5Br + CH_2 = CH - CH_2 - ONa$
(2) $CH_2 = CH - CH_2 - Br + C_6H_5ONa$
(3) $C_6H_5 - CH = CH - Br + CH_3 - ONa$
(4) $CH_2 = CH - Br + C_6H_5 - CH_2 - ONa$

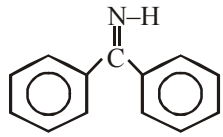
50.  + C_6H_5MgBr

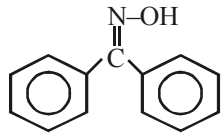
$$\xrightarrow{\text{Ether}} A \xrightarrow{H_3O^+} B$$

The final product (B) in above reaction is

(1) 

(2) 

(3) 

(4) 

Section - B

51. If radiation corresponding to second line of "Balmer series" of Li^{2+} ion, knocked out electron from first excited state of H-atom, then find the kinetic energy of ejected electron.

52. What is the sum of number of sigma (σ) and pi (π) bonds present in sulphuric acid molecule ?

53. Copper crystallises in *fcc* with a unit length of 361pm. What is the radius of copper atom ?

54. To find the standard potential of M^{3+}/M electrode, the following cell is constituted : Pt/ $M/M^{3+}(0.001 \text{ mol } L^{-1})/Ag^{+}(0.01 \text{ mol } L^{-1})/Ag$

The emf of the cell is found to be 0.421 volt at 298 K calculate the standard potential of half reaction

$M^{3+} + 3e^{-} \longrightarrow M$ at 298 K

(Given $E^{\circ}_{Ag^{+}/Ag}$ at 298 K = 0.80 Volt)

55. The number of π electrons present in 6.4 g of calcium carbide is $x \times N_A$ where (N_A = Avagadro's number). Find the value of x.

56. Two elements A & B form compounds having molecular formulae AB_2 and AB_4. When dissolved in 20.0 g of benzene 1.00 g of AB_2 lowers f.p. by 2.3 °C whereas 1.00 g of AB_4 lowers f.p. by 1.3 °C. The molal depression constant for benzene in 1000 g is 5.1. Calculate the sum of atomic masses of A and B.

57. An element (atomic mass =100 g/mol) having *bcc* structure has unit cell edge 400 pm. Calculate the density (in g/cm^3) of the element.

58. A substance "A" decomposes in solution following the first order kinetics. Flask I contains 1 L of 1 M solution of A and flask II contains 100 mL of 0.6 M solution. After 8 hr, the concentration, of A in flask I becomes 0.25 M. What will be the time for concentration of A in flask II to become 0.3 M?

59. Concentration of NH_4Cl and NH_4OH in a buffer solution is in the ratio of 1 : 1, K_b for NH_4OH is 10^{-10}. Calculate the pH of the buffer.

60. At 5×10^5 bar pressure and density of diamond and graphite are 3 g/cc and 2 g/cc respectively, at certain temperature 'T'. Find the value of $\Delta U - \Delta H$ for the conversion of 1 mole of graphite to 1 mole of diamond at temperature 'T'.

MATHEMATICS

Section - A

61. Which of the following statements is false?
(1) The length of sub-tangent to the curve $x^2y^2 = 16a^4$ at the point $(-2a, 2a)$ is $2a$.
(2) $x + y = 3$ is a normal to the curve $x^2 = 4y$
(3) Curves $y = -4x^2$ and $y = e^{-x/2}$ are orthogonal.
(4) If $a \in (-1, 0)$, then tangent at each point of the curve $y = \frac{2}{3}x^3 - 2ax^2 + 2x + 5$ makes an acute angle with the positive direction of x-axis.

62. If $\int \frac{(x^2-1)dx}{(x^4+3x^2+1)\tan^{-1}\left(\frac{x^2+1}{x}\right)} = \log|\tan^{-1} f(x)| + C$, then
(1) $f(x) = x^2 + 1$ (2) $f(x) = \frac{x^2+1}{2x}$
(3) $f(x) = \frac{x^2+1}{x}$ (4) $f(x) = \frac{1}{2}(x^2+1)$

63. If $P(\theta)$ and $Q\left(\frac{\pi}{2}+\theta\right)$ are two points on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, then locus of the mid- point of PQ is
(1) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=\frac{1}{2}$ (2) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=4$
(3) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=2$ (4) None of these

64. Area included between $y = \frac{x^2}{4a}$ and $y = \frac{8a^3}{x^2+4a^2}$ is
(1) $\frac{a^2}{3}(6\pi - 4)$ (2) $\frac{a^2}{3}(4\pi + 3)$
(3) $\frac{a^2}{3}(8\pi + 3)$ (4) None of these

65. Given that $\vec{a}$ is $\perp$ to $\vec{b}$ and p is a non zero scalar if $p\vec{r} + (\vec{r}.\vec{b})\vec{a} = \vec{c}$ then $\vec{r}$ equals
(1) $\vec{c}/p - \left[(\vec{b}.\vec{c})\vec{a}\right]/p^2$
(2) $\vec{a}/p - \left[(\vec{c}.\vec{a})\vec{b}\right]/p^2$
(3) $\vec{b}/p - \left[(\vec{a}.\vec{b})\vec{c}\right]/p^2$
(4) None of these

66. The domain of the function $f(x) = \sqrt{x - \sqrt{1-x^2}}$ is
(1) $\left[-1, -\frac{1}{\sqrt{2}}\right] \cup \left[\frac{1}{\sqrt{2}}, 1\right]$
(2) $[-1, 1]$
(3) $\left(-\infty, -\frac{1}{2}\right] \cup \left[\frac{1}{\sqrt{2}}, +\infty\right)$
(4) $\left[\frac{1}{\sqrt{2}}, 1\right]$

67. Which of the following is an empty set?
(1) The set of prime numbers which are even
(2) The solution set of the equation $\frac{2(2x+3)}{x+1} - \frac{2}{x+1} + 3 = 0;\ x \in \mathbf{R}$
(3) $(A \times B) \cap (B \times A)$, where A and B are disjoint.
(4) The set of real which satisfy $x^2 + ix + i - 1 = 0$

68. $f(x)=x^2[x]$

(1) increases in (0,1) (2) decreases in (0 , 1)

(3) increases in (–1,0) (4) None of these

69. If $y=e^x+\sin x$, then d^2x/dy^2 is equal to

(1) $e^x-\sin x$

(2) $-(e^x+\cos x)^{-2}$

(3) $-(e^x-\sin x)\ (e^x+\cos x)^{-2}$

(4) $(\sin x-e^x)(\cos x+e^x)^{-3}$

70. The sum to n terms of the series

$\frac{1}{2}+\frac{3}{4}+\frac{7}{8}+\frac{15}{16}+\ldots\ldots\ldots\ldots$ is

(1) $n-1-2^{-n}$ (2) 1

(3) $n-1+2^{-n}$ (4) $1+2^{-n}$

71. If $f: R\to R$ be a function defined by $f(x)=4x^3-7$. Then

(1) f is one-one -into (2) f is many-one - into

(3) f is many-one onto (4) f is bijective

72. Let $f:\left[-\frac{\pi}{3},\frac{2\pi}{3}\right]\to[0,4]$ be a function defined as

$f(x)=\sqrt{3}\sin x-\cos x+2$. Then $f^{-1}(x)$ is given by

(1) $\sin^{-1}\left(\frac{x-2}{2}\right)-\frac{\pi}{6}$

(2) $\sin^{-1}\left(\frac{x-2}{2}\right)+\frac{\pi}{6}$

(3) $\frac{2\pi}{3}+\cos^{-1}\left(\frac{x-2}{2}\right)$

(4) None of these

73. The roots α and β of the quadratic equation $px^2+qx+r=0$ are real and of opposite signs. The roots of $\alpha(x-\beta)^2+\beta(x-\alpha)^2=0$ are

(1) positive (2) negative

(3) of opposite signs (4) non-real

74. The equation $k\sin\theta+\cos 2\theta=2k-7$ possesses a solution if :

(1) $2\le k\le 6$ (2) $k>2$

(3) $k>6$ (4) $k<2$

75. If $y=\sin^{-1}\frac{2x}{1+x^2}$, then which of the following is not correct?

(1) $\frac{dy}{dx}=\frac{2}{1+x^2}$ for $|x|<1$

(2) $\frac{dy}{dx}=\frac{-2}{1+x^2}$ for $|x|>1$

(3) $\frac{dy}{dx}=2$ for $x=-1$

(4) $\frac{dy}{dx}$ does not exist at $|x|=1$

76. If the slope of the tangent at (x, y) to a curve passing through $\left(1,\frac{\pi}{4}\right)$ is given by $\frac{y}{x}-\cos^2\left(\frac{y}{x}\right)$, then the equation of the curve is

(1) $y=\tan^{-1}\log(e/x)$

(2) $y=e^{1+\cot(y/x)}$

(3) $y=x\tan^{-1}\log(e/x)$

(4) $y=e^{1+\tan(y/x)}$

77. If $y=\tan^{-1}\left(\frac{2^x}{1+2^{2x+1}}\right)$, then $\frac{dy}{dx}$ at $x=0$ is :

(1) $-\frac{3}{5}\log 2$ (2) $\frac{2}{5}\log 2$

(3) $-\frac{3}{2}\log 2$ (4) None of these

78. If $\begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix} = 0$ and the vectors $\vec{A} = (1,a,a^2)$; $\vec{B} = (1, b, b^2)$; $\vec{C} = (1,c, c^2)$ are non-coplanar then the product abc =

(1) 0 (2) 1

(3) –1 (4) None

79. A and B are two independent witnesses (i.e. there is no collusion between them) in a case. The probability that A will speak the truth is x and the probability that B will speak the truth is y. A and B agree in a certain statement. The probability that the statement is true is

(1) $\frac{x-y}{x+y}$ (2) $\frac{xy}{1+x+y+xy}$

(3) $\frac{x-y}{1-x-y+2xy}$ (4) $\frac{xy}{1-x-y+2xy}$

80. Let x_1 and y_1 be real numbers. If z_1 and z_2 are complex numbers such that $|z_1| = |z_2| = 4$, then $|x_1z_1 - y_1z_2|^2 + |y_1z_1 + x_1z_2|^2 =$

(1) $32(x_1^2 + y_1^2)$ (2) $16(x_1^2 + y_1^2)$

(3) $4(x_1^2 + y_1^2)$ (4) $8\left(x_1^2 + y_1^2\right)$

Section - B

81. Let f be a composite function of x defined by $f(u) = \frac{1}{u^2+u-2}, u(x) = \frac{1}{x-1}$. Then the number of points x where f is discontinuous is _____.

82. If the function $f(x) = Pe^{2x} + Qe^{x} + Rx$ satisfies the conditions $f(0) = -1, f'(\log 2) = 31$ and $\int_0^{\log 4} (f(x) - Rx)dx = \frac{39}{2}$, then valve of P + Q + R is _____.

83. The sum of 10 items is 12 and sum of their squares is 18, then standard deviation will be _____.

84. If the system of equations :

$p^3x + (p+1)^3 y = (p+2)^3$

$px + (p+1)y = p+2$

$x + y = 1$

is consistent then valve of |P| is _____.

85. If $f(x) = 3x^{10} - 7x^8 + 5x^6 - 21x^3 + 3x^2 - 7$, then $\lim_{\alpha \to 0} \frac{f(1-\alpha) - f(1)}{\alpha^3 + 3\alpha}$ is _____.

86. A survey of 500 television viewers produced the following information, 285 watch football, 195 watch hockey, 115 watch basket-ball, 45 watch football and basket ball, 70 watch football and hockey, 50 watch hockey and basket ball, 50 do not watch any of the three games. The number of viewers, who watch exactly one of the three games are _____.

87. The value of $|x|$ for which $\sin(\cot^{-1}(1+x)) = \cos(\tan^{-1} x)$ is _____.

88. The value of $\lim_{x \to 0} \frac{1-\cos^3 x}{x \sin x \cos x}$ is _____.

89. If $f(x) = \frac{1}{1-x}$, the number of points of discontinuity of $f\{f[f(x)]\}$ is _____.

90. The number of roots of equation $\cos x + \cos 2x + \cos 3x = 0$ is $(0 \le x \le 2\pi)$ _____.

Space for Rough Work